공간적 경계와 랜드마크가 노년층의 일화기억 및 공간 인지에 미치는 영향

The Impact of Boundaries and Landmarks on Episodic Memory and Spatial Cognition in the Elderly

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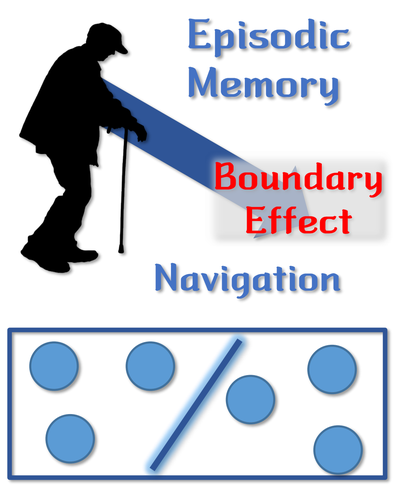
INTRODUCTION



Global Trend Aging

- Studying how the elderly perceive and remember space is critical for designing supportive environments.

Declining Space Cognition as Aging

- Research has shown that as we age, our ability to perceive space and remember it declines.

Boundary effect

- Recognizing space using boundaries helped with episodic memory related to space.

OBJECTIVES

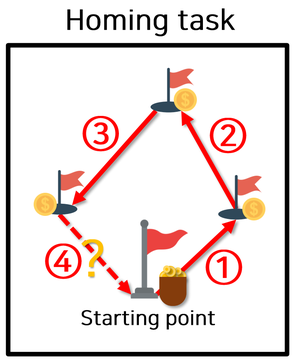
- Design an optimal task environment that can clearly differentiate the effects of boundary and landmark
- Understand how boundary influences spatial factors of episodic memory in the elderly, through the analysis of experimental results.

METHODS

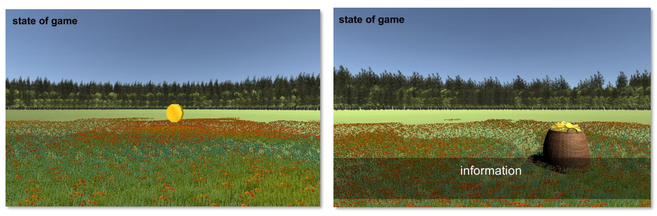
1. Path Integration Task

- Test the ability to calculate position relative to the starting point by summing up distances and directions traveled

Homing task

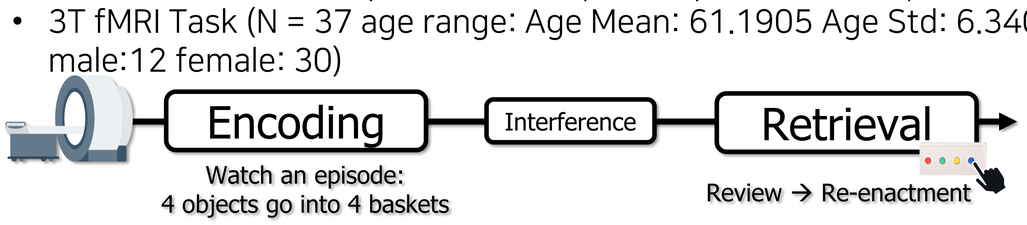


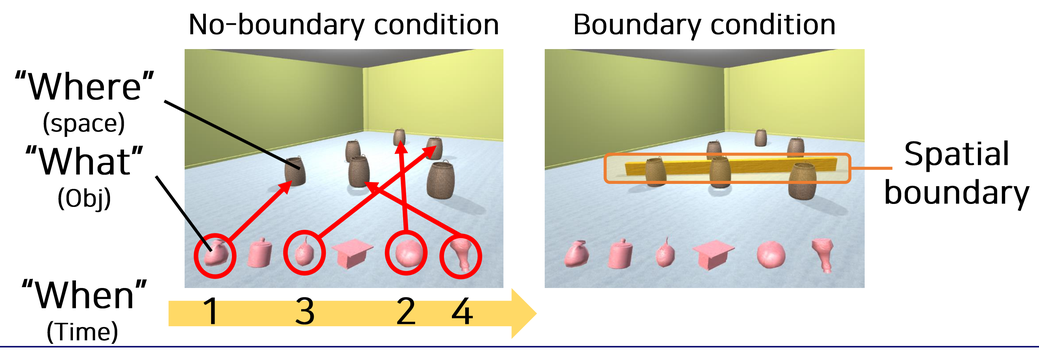
- Participants:
  - MCI (Mild Cognitive Impairment)
  - AD (Alzheimer's Disease) patients
- Example scene: first person point of view



2. Episodic Memory with Boundary Task

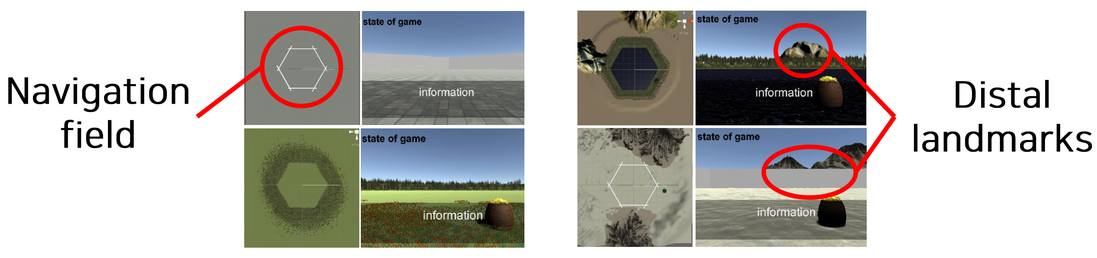
- To test the effect of spatial boundary on sequence memory
- 3T fMRI Task (N = 37 age range: Age Mean: 61.1905 Age Std: 6.3409 male:12 female: 30)





RESULTS

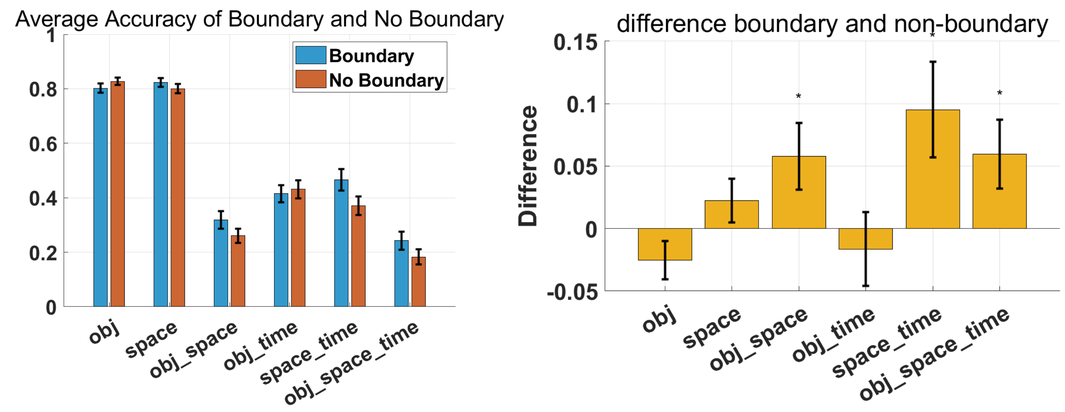
1. Path Integration Task: Making navigation environments



RESULTS

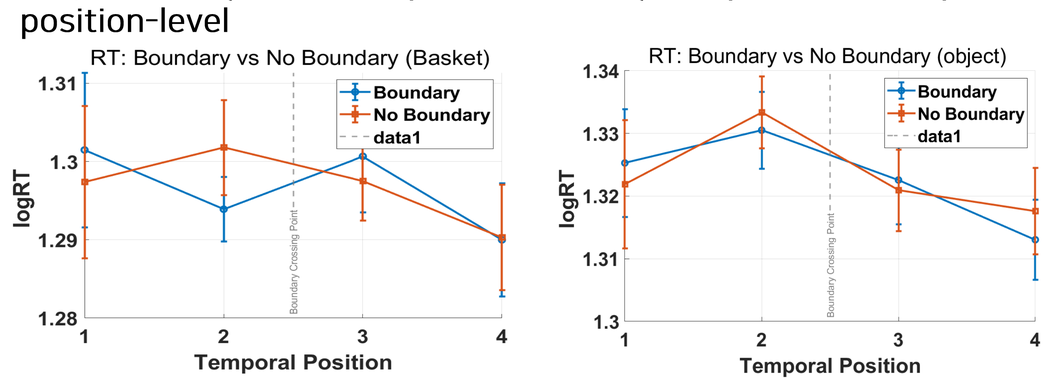
2. Episodic Memory with Boundary Task

2-1) boundary effect in episodic memory components: trial-level



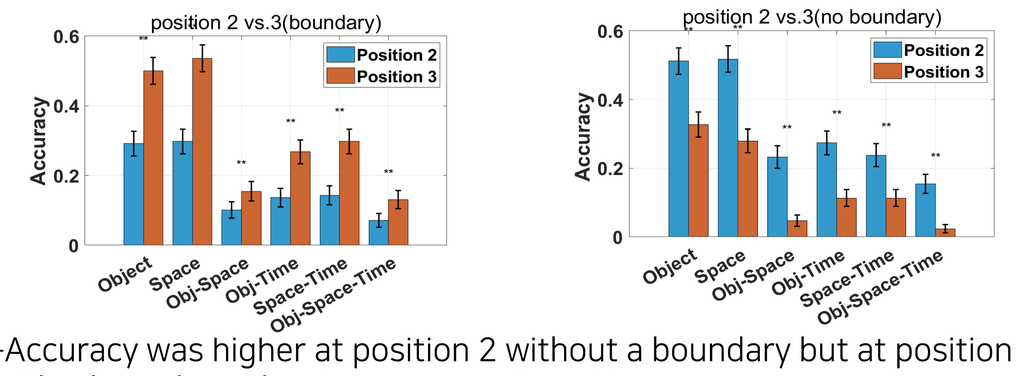
-Only fields involving space(basket) showed a significant positive effect.

2-2) boundary effect in episodic memory components: Temporal position-level



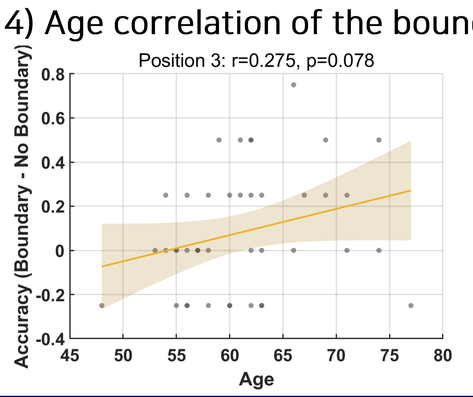
-RT decreased pre-boundary and increased post-boundary.

3) Pre boundary-post boundary accuracy analysis



-Accuracy was higher at position 2 without a boundary but at position 3 with a boundary, showing an opposite pattern.

4) Age correlation of the boundary effet



Accuracy differences grew with age, showing greater boundary influence on older participants.

CONCLUSIONS & FUTURE PLAN

- Boundaries significantly enhance the accuracy of spatial recall in episodic memory among the elderly.
- Furthermore, this positive effect increases with advancing age in participants.
- Future analyses will include results from younger participants and the type of boundary (vertical or horizontal).

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